

Fredson Silva de Souza Aguiar

☎ +55 21 99136-9523 — 🌐 fredsonaguiar.github.io — ✉ fredsonaguiar704@gmail.com — 📄 fredson-aguiar-704

Education

Université Sorbonne Paris Nord

M.Sc. (M2) in Mathematics of Data

GPA: 17.26 / 20.00

Île-de-France, France

2025 – 2026

Escola de Matemática Aplicada (FGV/EMAp)

M.Sc. in Applied Mathematics and Data Science

GPA: 9.25 / 10.00

Rio de Janeiro, Brazil

2023 – 2025

Escola de Matemática Aplicada (FGV/EMAp)

B.Sc. in Applied Mathematics

GPA: 9.10 / 10.00

Rio de Janeiro, Brazil

2018 – 2022

Technical Skills & Core Competencies

- **Core Fields:** Optimal Control Theory, Reinforcement Learning, Model Predictive Control (MPC), Dynamical Systems, Neural Networks, Numerical Analysis and Simulation.
- **Languages & Tools:** Python (PyTorch, CasADi, SciPy, NumPy, etc.), R, C++, LaTeX, Git.

Research & Professional Experience

Research Intern

LAGA – Laboratoire de Analyse, Géométrie et Applications

Project in Association with Safran & Polytechnique Montréal

Jul 2026 – Sep 2026

- Investigating machine learning applicability for predictive maintenance in aircraft systems; natural language models for retrieval, question-answering, and summarization, RAG and GraphRAG (see Py-MPREV).

Research Intern

FAU Erlangen-Nürnberg

Chair for Dynamics, Control and Numerics (Alexander von Humboldt Professorship)

Apr 2026 – Jul 2026

- Analyzed connections between classical Optimal Control and Reinforcement Learning algorithms through the lens of Linear Quadratic Regulator (LQR) problems (see Project Report).

NLP Intern

IBM Brazil

Natural Language Processing Team

Jun 2021 – Nov 2021

- Developed and implemented data repair, enriching and publication pipelines for the OpenWordnet, a relevant Natural Language dataset in Portuguese (see Py-OWNPT).

Quantitative Data Science Intern

Bank BOCOM BBM

Credit Analysis & Data Lake Team

Jan 2021 – Feb 2021

- Developed automation workflows for credit risk analysis data within the enterprise financial data lake, focused on the real estate sector.

Undergraduate Researcher (PICME/CNPq)

FGV/EMAp

Semantic Representations & Ontologies

Aug 2019 – Jul 2021

- Researched semantic representations, the use of logical knowledge bases in textual inference problems, and word embedding strategies for expanding lexical databases.

Publications & Preprints

- Aguiar, F., Aronna, M. S., Guglielmi, R., Laiate, B., & Molter, A. (2026). **Host-Parasitoid Dynamics and Biological Control of the Sugarcane Borer**. *arXiv preprint arXiv:2607.12878*.
- Morel, A., Oumar, M., Ali, C., Yazid, F., Fredson, A., Keltoum, K., Amadou, D., & Lacaille, J. (2026). **A Stochastic Event-Generation Simulator for Dynamic Risk Assessment and Maintenance Decision Support in Aircraft Systems**. *Proc. CM 2026*.
- Aguiar, F. S. S., & Aronna, M. S. (2025). **A Learning-Based Model Predictive Control Method for Epidemic Mitigation**. *X MACI 2025*.
- Oliveira, H. G., Aguiar, F. S. S., & Rademaker, A. (2021). **On the Utility of Word Embeddings for Enriching OpenWordNet-PT**. *Language, Data and Knowledge (LDK 2021)*, OASICS, Vol. 93, pp. 21:1–21:13.

Other Selected Projects

- **Master’s Monograph:** *A Learning-Based Model Predictive Control Framework with a State Predictor for Epidemic Mitigation under Dynamics and Parametric Uncertainty.* (FGV/EMAp) [Link]
- **Internship Report:** *From Optimal Control to Reinforcement Learning: A Motivating Example from the Perspective of Linear Quadratic Regulators.* (FAU-MoD) [Link]
- **Condor Fellowship Project:** *Inverse Optimal Control and Model Predictive Control for Autonomous Decision Making Under Timely Distributional Shift.* [PDF]

Teaching & Leadership Activities

Teaching Assistant <i>Graduate & Undergraduate Level Courses</i>	Escola de Matemática Aplicada (FGV/EMAp) 2020 – 2024
• Graduate (M.Sc. / Ph.D.): Introduction to Functional Analysis, Qualitative Theory of ODEs, Multivariate Analysis.	
• Undergraduate: Real Analysis, Partial Differential Equations, Complex Calculus, Vector Calculus, Discrete Mathematics.	
Graduate Student Seminar Coordinator Organized research dissemination talks and academic seminars	FGV/SEPEMAP Aug 2023 – Aug 2025
Academic Tutor & Volunteer Educator Tutored mathematically talented students for selective programs	FGV/CDMC & Amplia FGV Jan 2019 – Nov 2024

Honors & Selective Events

• The Len Gelman Award (BINDT CM2026): Best Paper in Early Career Stage	June 2026
• 12th Heidelberg Laureate Forum (12HLF): Selected Young Researcher	Sep 2025
• XIV BrazOpt: Theoretical and Computational Continuous Optimization	Mar 2024
• EAGx Latin America 2024: Effective Altruism, AI Safety, and AI Governance	Feb 2024
• Condor Camp 2024: Bootcamp in Career Acceleration, AI Safety, and AI Governance	Feb 2024
• Tropical Probabilistic AI School 2024: Probabilistic Machine Learning and AI	Jan 2024 – Feb 2024